

The Aging Kidney and Acute Kidney Injury

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Abstract

Aging is unavoidable and driven, at least in part, by the biologic process termed senescence. Senescence is characterized by cell cycle arrest incurred by one or more stressors wherein there is cellular upregulation of cyclin-dependent kinase inhibitors, p16^{Ink4a} and/or p21^{Cip1}. Senescent cells are metabolically active and resistant to apoptosis, have increased β -galactosidase activity, and express a senescence-associated secretory phenotype (SASP). The SASP produces an array of inflammatory, cytotoxic, fibrotic, and vasoactive factors. Senescence is a diffusible process as, largely through the SASP, a focus of senescent cells can recruit adjacent cells and cells in distant organs, thereby expanding its scope way beyond its original focus. Senescence predisposes not only to chronic diseases (including CKD) but also acute processes (including AKI). This review provides evidence that age increases the risk of human AKI and summarizes clinical factors that predispose older individuals to AKI. It provides an overview of the biology of senescence, and it discusses the role of intrarenal senescent cells in the pathogenesis of AKI, the significance of the extrarenal milieu and distant organs in predisposing to AKI with age, heme as a prosenescent species, and the role of senescence in the AKI to CKD transition. The risk for AKI with aging, in large part, reflects the fact that virtually all processes implicated in AKI, irrespective of age, occur as the kidney ages. Processes discussed include senescent cells, vascular responses, diabetes, impaired NAD⁺ content, sirtuin expression, mitochondrial dysfunction, impaired autophagy, epigenetic changes, telomere shortening, impaired nephroprotectant expression, inflammaging, and immunosenescence. The review concludes by discussing the basis for senolytics (agents that kill senescent cells), senomorphics (agents that block SASP factors), and other aspects of senotherapies.

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It is a truism—but an elemental and needed introductory one—that human populations are aging. In 5 years, the US Census Bureau estimates that all baby boomers will be at retirement age and comprise 20% of the US population.¹ This watershed year in US demographics will also observe an aging global population with one in six people older than 60 years, the latter totaling 1.4 billion individuals worldwide.²

Aging, in its inescapable course, predisposes to chronic diseases (including CKD) and AKI (Figure 1 and Table 1A).^{3–9}

Evidence That Age Increases the Risk of and Worsens Human AKI

United States Renal Data System data in 2016 demonstrate that, compared with such rates in individuals aged 66–69 years, the rate of AKI more than doubled for those 80–84 years and more than quadrupled for those older than 85 years¹⁰; and that the likelihood of recurrent AKI within 1 year is higher in older compared with younger patients.¹⁰ Additionally, recovery of kidney function after

AKI is impaired in patients 65 years and older.¹¹ The age-dependent risk of AKI is higher when there is concomitant CKD and diabetes; 20% of hospitalized Medicare patients with underlying CKD and diabetes develop AKI each year.¹⁰ Earlier studies demonstrate that dialysis-requiring AKI increased by some 10% each year from 2000 to 2009, with the incidence rate progressively increasing with age.¹² However, more recently, the percentage of hospitalizations with AKI necessitating dialysis in older individuals has plateaued or even decreased, with such percentages decreasing numerically from the age range 66–69 years to those 85 years and older.¹³

Risk Factors for AKI in Older Patients

Factors predisposing to clinical AKI with age are listed in Table 1A. Alterations in the aged kidney themselves predispose to AKI because these alterations impair responses to, or recovery from, acute stressors (Table 1B).¹⁴ The kidney in CKD is differentiated from the kidney in unhealthy and healthy aging by the relative greater loss of nephrons in CKD,¹⁴ a greater burden of senescent

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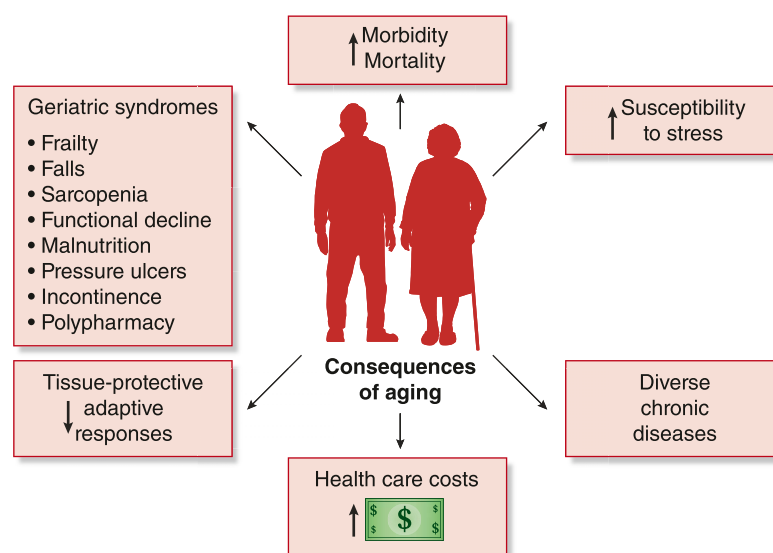


Figure 1. Consequences and complications of aging. Some, but not all, geriatric syndromes are listed.

cells,¹⁵ more prominent inflammaging,¹⁶ and tubulointerstitial disease in CKD¹⁷ (Figure 2).

Overview of the Biology of Aging and Cellular Senescence

Aging was previously considered a consequence of the intrinsic limitations in the replicative capacity of mammalian cells, the “Hayflick limit.”¹⁸ Aging occurred because

cells, lost during life’s “wear and tear,” could not be repopulated because of set limits on cell division. In spaces in which parenchymal cells had died and disappeared, progressive fibrosis ensued.

The current concept of aging is founded upon cellular senescence, a biologic process driven by cell cycle-arrested, senescent cells, with their characteristics listed in Table 2.^{19–22} Assorted stressors (ischemic, genotoxic, nephrotoxic, hypoxic, oxidative, inflammatory) upregulate cyclin-dependent kinase inhibitors p16^{Ink4a} and/or p21^{Cip1} with attendant interruption in cell cycle progression. It should be noted, however, that p16^{Ink4a} and p21^{Cip1} may affect processes other than the cell cycle.^{23,24}

Senescent cells injure tissues through the senescence-associated secretory phenotype (SASP). The SASP elaborates diverse cytokines; inflammatory, fibrogenic, pro-oxidant, proapoptotic, cytotoxic, and reactive nitrogen species; several proteases; and vasoconstrictive agonists. Through the SASP, a focus of senescent cells can commandeer once normal neighboring and distant cells into senescence: senescence is thus a diffusible process. Through the SASP, senescent cells often escape immunologic surveillance and removal (Figure 3).

Senescence informs major aging paradigms proposed over the past decade. The Geroscience Hypothesis of Aging enumerated seven pillars of aging and emphasized their connectedness such that age-related chronic diseases ensued.²⁵ The Hallmarks of Aging listed nine pathogenetic processes,²⁶ which were expanded to 14 in 2022 (Table 3).²⁷ The Unitary Theory of Fundamental Aging Mechanisms underscored the interaction of the pillars/hallmarks of aging, and that because of such interdigitation, strategies that interrupt any one of these pillars/hallmarks may also exert inhibitory effects on others.²⁸ The interruption of senescence by whatever approach would thus be anticipated to have broad-based salutary effects.

Senolytics, agents that selectively kill senescent cells, were discovered in 2016 following analyses that revealed upregulation of survival pathways in senescent cells and

Table 1. The aging kidney and AKI: (A) increased risk of AKI in the older population and (B) attributes of the aging kidney

(A) Risk Factors Contributing to AKI in the Older Population	
1	Increased likelihood of exposure to nephrotoxic medications
2	Higher frequency of surgical procedures, interventions, and need for general anesthesia
3	Greater number of diagnostic imaging procedures requiring potentially nephrotoxic contrast dyes
4	Elevated cancer risk and exposure to nephrotoxic cancer therapies and irradiation
5	Susceptibility to chronic diseases leading to higher risk for AKI
	• Type 2 diabetes • Cardiovascular diseases • CKD • Renovascular diseases • Hypertension • Certain glomerulonephritides • Heart failure • Obesity
(B) Attributes of the Aging Kidney	
Functional Changes	Structural Changes
↓ GFR ↓ Renal blood flow ↓ Nephron number ↓ Mitochondrial function - Inflammaging - SASP factors	- Cortical thinning - Arterio/arteriosclerosis - Glomerulosclerosis - Interstitial fibrosis - Tubular atrophy ↑ Senescent cells
SASP, senescence-associated secretory phenotype.	

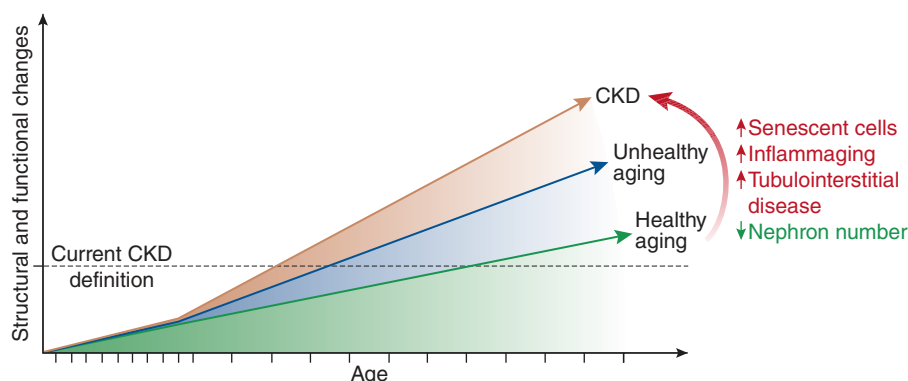


Figure 2. Differentiation of the kidney in CKD from the kidney in unhealthy and healthy aging. As the kidney deteriorates from healthy aging to unhealthy aging to CKD, there is a progressive increase in the population of senescent cells, the severity of inflammation, and tubulointerstitial disease, with a progressive decrease in nephron number. Unhealthy aging reflects the presence of comorbidities, such as cardiovascular diseases, metabolic diseases, and cancer, among other diseases. Modified from ref. 14, with permission.

that the silencing of critical nodes in these pathways selectively killed senescent cells.²⁹ Such survival pathways include the Bcl-2 family, and inhibitors of Bcl-2, Bcl-xL, and Bcl-w, such as ABT-263, are senolytic (see next section). Upregulated survival pathways in senescent cells explain why senescent cells withstand the inimical SASP factors they produce. Drug screening strategies then identified already approved drugs or natural compounds that kill senescent cells *in vitro* and *in vivo*. Senolytics delay chronic diseases and improve longevity in numerous disease models.^{19–22} Senolytics are also effective in humans *in vivo*.³⁰

Evidence supporting senescence as a cause of aging and chronic disease also comes from studies using *INK-ATTAC* mice wherein p16^{Ink4a}-expressing cells can be deleted. Such deletion interrupts aging in a progeroid model³¹ and, in wild-type mice, increases longevity, reduces age-related organ injury (including kidney), and delays carcinogenesis.³²

Senescence and AKI

The Role of Intrarenal Senescent Cells

Beginning with the demonstration that the renal senescent cell burden correlates with kidney dysfunction, seminal studies established the critical contribution of

senescent cells in murine AKI.³³ Aged mice compared with younger mice, subjected to ischemia-reperfusion injury (IRI), have a greater expression of senescence markers, SASP factors, and kidney fibrosis. When the senolytic, ABT-263, was administered to aged mice,

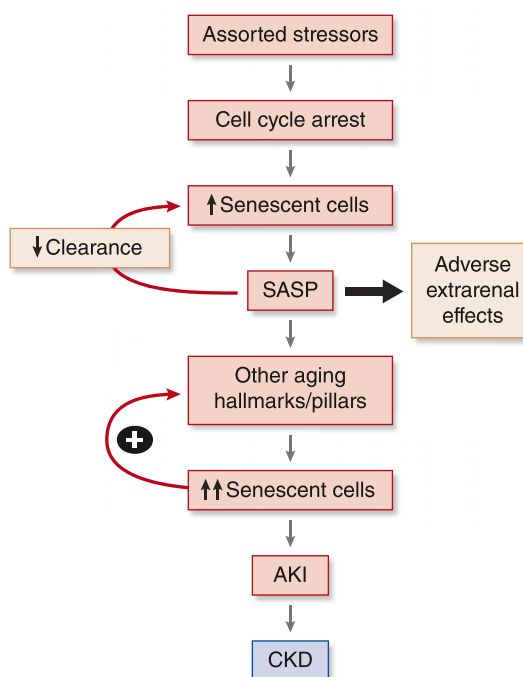


Figure 3. Schema outlining pathways whereby acute stresses to the kidney culminate in AKI. Such stressors lead to cell cycle arrest, the emergence of senescent cells, and the appearance of SASP factors. SASP factors may prevent the clearance of senescent cells by immune mechanisms. The SASP may exert adverse effects both locally as well as on distant organs. Senescent cells and SASP factors may recruit other hallmarks/pillars of aging. It should be pointed out that while senescent cells can predispose to AKI, it is unclear/unknown if senescent cells *per se* cause AKI in the absence of other insults. SASP, senescence-associated secretory phenotype.

Table 2. Characteristics of senescent cells

- 1 Age-related accumulation in tissues
- 2 Cell cycle arrest due to upregulation of p16, p21, or both
- 3 Cells are viable, metabolically active, and often enlarged
- 4 Characteristic biochemical features
 - ↑ Senescence-associated β galactosidase activity
 - Loss of lamin B1
 - Leaky lysosomes
 - ↑ γ -H2A.X
- 5 Resistance to apoptosis (programmed cell death)
- 6 Exhibit SASP

SASP, senescence-associated secretory phenotype.

Table 3. New hallmarks of aging^a

Cellular senescence
Inflammation
Telomere attrition
Mitochondrial dysfunction
Altered intracellular communication
Epigenetic alterations
Genomic instability
Splicing dysregulation
Altered mechanical properties
Loss of proteostasis
Deregulated nutrient-sensing
Microbiome disturbance
Compromised autophagy
Stem cell exhaustion

^aAs listed in ref. 27.

senescent cells were cleared, and following IRI, ABT-263-treated mice exhibited increased GFR, cell proliferation, and renal regeneration, and reduced matrix expansion.³³ These findings were complemented by studies in which young mice were subjected to irradiation to increase the renal burden of senescent cells. When these mice were treated with ABT-263 before IRI, it was again observed that the senescent cell burden, kidney dysfunction, and histologic injury were attenuated, whereas cell proliferation was increased. Thus, senescent cells, in the naturally aged kidney, or recruited by irradiation in the kidney of young mice, exacerbate IRI and impair recovery from IRI.³³

Prior observations support this paradigm. AKI is less severe in mice with homozygous deficiency of p16^{Ink4a} compared with wild-type mice.^{34,35} INK4a/ARF^{-/-} mice, which are deficient in p16^{Ink4a} (also p19), after IRI, compared with wild-type mice, display improved kidney function, less apoptosis, and increase cellular proliferation.³⁶ Kidneys from INK4a/ARF^{-/-} mice, transplanted in wild-type mice, exhibit increased proliferating tubular cells and less interstitial fibrosis, and these recipients have

greater survival.³⁴ Previous studies also underscore the importance of proliferative processes in AKI recovery; for example, in aged mice, proliferation is impaired either because of increased expression of inhibitors³⁷ or decreased expression of stimulators of cell proliferation,³⁸ either process impairing AKI recovery.^{37,38}

The Role of the Extrarenal Milieu

The systemic milieu in AKI includes abundant cytokines and other bioactive species. Such species exert adverse effects of AKI on extrarenal tissues and distant organs. Dysfunction of distant vital organs and circulating bioactive substances, in turn, feedback to the kidney and worsen AKI; bidirectional processes, from and to the kidney, are thus instigated in AKI.³⁹

Parabiosis conjoins and thus shares the circulation between mice. This technique was used to explore whether the extrarenal milieu—from old or young mice—influences outcomes from IRI in old mice.⁴⁰ Old-old parabiosis mice had comparably severe IRI compared with old mice after IRI. However, following IRI, old-young parabiosis mice compared with old-old parabiosis mice exhibited less severe AKI.⁴⁰ Thus, the age-dependent increased AKI sensitivity reflects not just age-dependent processes within the kidney but also on age-dependent processes in the systemic milieu and distant organs.

What is the explanation for this finding? We suggest that the adverse effects of an “old” extrarenal milieu reflects, in part, the increased susceptibility of “old” extrarenal organs to SASP factors from the injured kidney (Figure 4). This would impair the function of vital organs, which, itself, would worsen AKI. SASP factors from senescent cells in AKI, transmitted through the circulation to distant organs, would recruit senescent cells in distant tissues, which would contribute their own SASP factors back into the circulation; this would augment SASP-driven adverse effects on the injured kidney. These concepts are fully substantiated in the senescence literature in other tissues.^{19–22,28}

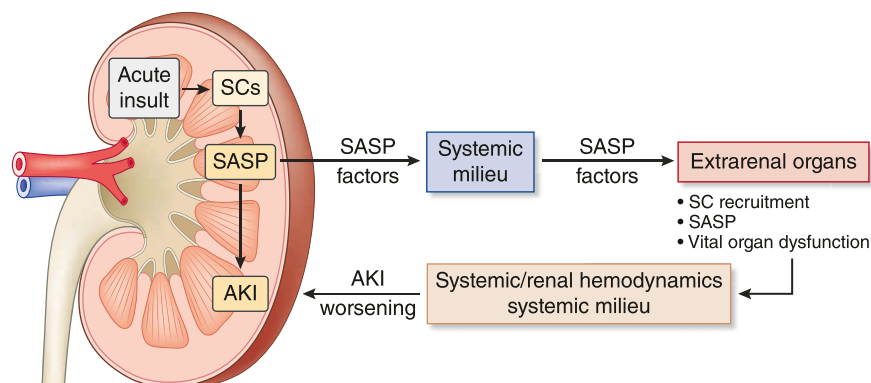


Figure 4. Bidirectional effects involving the kidney and extrarenal organs in AKI. SASP factors not only contribute to AKI but also exert distant effects on extrarenal organs where senescent cells are recruited with the attendant production of SASP factors. This could impair the function of distant vital organs which, in turn, through systemic and renal hemodynamics, the systemic milieu, and the renal circulation may worsen AKI. It should be pointed out that while senescent cells can predispose to AKI, it is unclear/unknown if senescent cells *per se* cause AKI in the absence of other insults. SC, senescent cell.

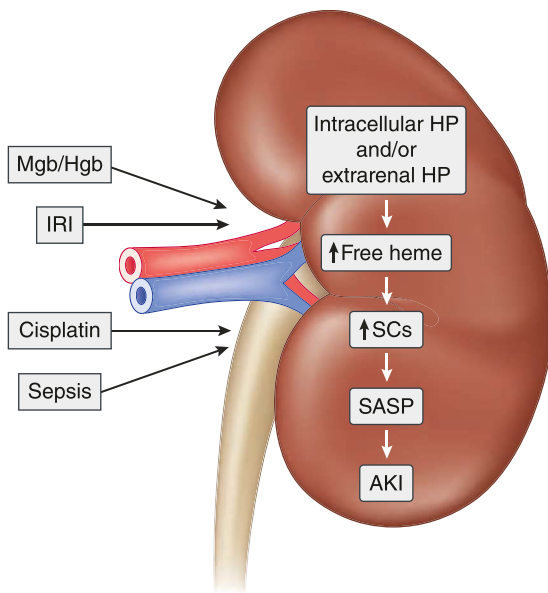


Figure 5. Free heme content in the kidney as an instigator of senescence in AKI. In IRI and cisplatin AKI, there is increased intrarenal content of free heme arising from destabilized HPs. In HP-AKI, either Mgb or hemoglobin is released into the circulation and incorporated into the kidney. In sepsis-associated AKI, there are increased circulating levels of free hemoglobin, which also is incorporated into the kidney giving rise to increased heme content in the kidney. To aid in presenting this concept, this figure is markedly simplified, in particular, by omitting other major pathogenetic pathways in these various forms of AKI and by omitting myriad effects of HPs other than their prosenescent effects. Hgb, hemoglobin; HP, heme protein; HP-AKI, heme protein-induced AKI; IRI, ischemia-reperfusion injury; Mgb, myoglobin.

Heme Protein–Induced AKI

Instigated by myoglobin or hemoglobin, heme protein–induced AKI (HP-AKI) exhibits ischemia, cytotoxicity, cell death, and inflammation.⁴¹ To the extent that such acute stressors instigate senescence, senescence will occur in HP-AKI; indeed, age worsens the severity of HP-AKI.⁴² Unique to HP-AKI is the large heme burden imposed on the kidney,⁴³ and heme itself, as shown by our studies, induces senescence.^{44–46} In HP-AKI *in vivo*, multiple markers including induction of p16^{Ink4a} appear within hours of HP-AKI, and genetic manipulations that increase renal heme content increase expression of p16^{Ink4a} and worsen HP-AKI. Moreover, using *INK-ATTAC* mice, we demonstrate that deletion of p16^{Ink4a}-expressing cells improved kidney function in HP-AKI.⁴⁴ Thus, the pathogenesis of HP-AKI involves senescence and p16^{Ink4a} induction, the latter driven, in part, by hemoglobin, myoglobin, and/or heme. Heme, however, is also elevated in the kidney in other AKI models including IRI,⁴⁷ cisplatin-induced AKI,⁴⁸ and sepsis-induced AKI.^{41,49} In IRI and cisplatin-AKI, heme is thought to arise from destabilized and/or degraded intracellular heme proteins,^{43,50–52} while sepsis AKI is associated with increased circulating levels of hemoglobin,⁴⁹ the latter incorporated in the kidney (Figure 5). Heme, in essence, may be an intracellular toxicant irrespective of the cause of AKI,⁴³ and, in part, through its prosenescent effects.^{44–46}

AKI to CKD Transition

Studies based on the administration of senolytics or p16^{Ink4a} knockdown demonstrate the role of senescence in the AKI to CKD transition.^{53–55} The AKI to CKD transition reflects maladaptive responses that promote vascular rarefaction, compromise repair, and incite inflammatory and fibrogenic responses.^{56,57} Notably, bypassing G2/M arrest by a p53 inhibitor prevents fibrosis after unilateral IRI.⁵⁸

Overlapping Senescence-Related and AKI-Related Pathways

Renal aging is attended by assorted processes also implicated in AKI, irrespective of the age at which AKI occurs. Such overlapping processes may underlie the age-dependent risk of AKI (Figure 6).

Accumulating Senescent Cells

Senescent cells accumulate as the kidney ages,¹⁵ and through SASP factors, senescence not only spreads locally and distantly but it also sensitizes otherwise normal neighboring cells to AKI. Thus, senescence and AKI interact in at least two major ways: first, senescent cells and their SASP factors within the aged kidney predispose to AKI; and second, acute insults to the kidney, irrespective of age, can instigate a senescence program that engenders the appearance of senescent cells.

Tubulointerstitial, glomerular, and vascular cells can all senesce and exert adverse paracrine effects.⁵⁹ For example, senescent glomerular endothelial cells produce PAI-1, which causes podocyte loss, thereby eliciting glomerulosclerosis.⁶⁰ With their high metabolism, tubular cells are stress-sensitive but can regenerate, unlike podocytes. Aging tubular cells exhibit tubuloinflammation (including MHC-II gene upregulation), which sensitizes them to acute insults.⁶¹ As renal compartments communicate with one another,¹⁷ senescent cells in one compartment, *via* SASP factors, can recruit senescent cells in other compartments.

Altered Vascular Responses

Aged kidneys exhibit increased sensitivity to IRI, caused, in part, by decreased nitric oxide availability, as the latter is scavenged by superoxide anion⁶²; the aged kidney generates increased amounts of superoxide anion *via* NOX isoforms.⁶³ The aged kidney exhibits impaired vasodilatory responses,⁶⁴ increased vasoconstrictive responses,⁶⁵ heightened stress-induced vasoconstriction,⁶⁶ and impaired autoregulation.⁶⁷ All these age-dependent, altered vascular responses may underlie increased sensitivity to AKI in older patients.

In the aging kidney⁶⁸ and in surviving nephrons,⁶⁹ the glomerular capillary hydrostatic pressure increases, and agents that retard CKD progression—inhibitors of the renin-angiotensin-aldosterone system and sodium-glucose cotransporter 2—reduce glomerular capillary pressure. Interestingly, pressure-induced, cellular stretching may cause senescence,⁷⁰ and, in this regard, these inhibitors exert antisenescent effects.^{71,72}

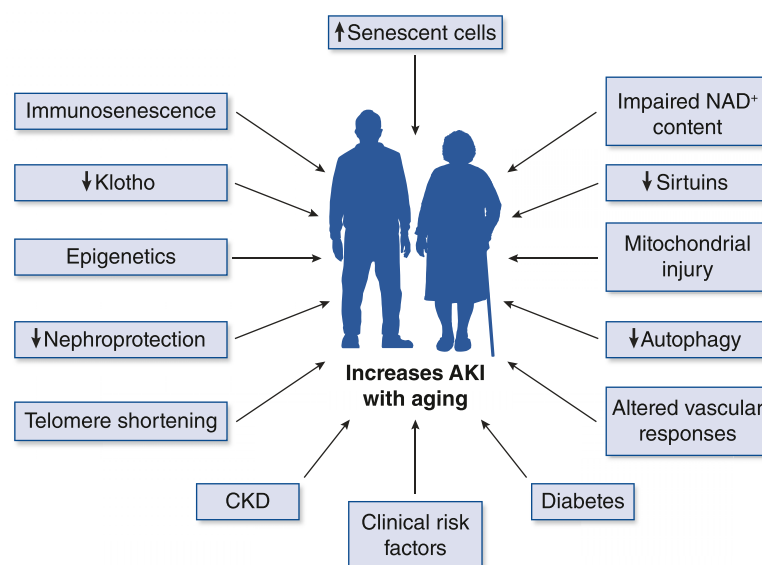


Figure 6. Overlapping processes/pathways that occur in the aging kidney and may also contribute to AKI.

Impaired NAD⁺ Availability

Aging is associated with decreased tissue NAD⁺ content.^{73–76} In preclinical models, boosting NAD⁺ mitigates age-related syndromes and susceptibility to injury. NAD⁺ and NADH are involved in oxidation-reduction reactions essential for intermediary metabolism and ATP generation.^{73–76} NAD⁺ is needed by dehydrogenases in glycolysis and the TCA cycle, becoming reduced to NADH; NADH donates electrons to complex 1 in the electron transport chain during oxidative phosphorylation, being oxidized to NAD⁺. NAD⁺ is also consumed by enzymes that convert NAD⁺ to nicotinamide (NAM).

In health, NAD⁺ consumption is balanced by commensurate NAD⁺ generation, the latter achieved by no less than three pathways^{73–76}: the salvage pathway, which generates NAD⁺ from NAM; the dietary tryptophan-driven *de novo* pathway; and the dietary niacin-driven Preiss–Handler pathway.

Seminal studies demonstrate the significance of impairment in the *de novo* NAD⁺ synthetic pathway in AKI.⁷⁷ First, in murine IRI, renal NAD⁺ content decreases, quinolinate (an intermediate in the *de novo* pathway) accumulates, and expression of quinolinate phosphoribosyl transferase (the enzyme that converts quinolinate ultimately to NAD⁺) decreases. Second, mice deficient in quinolinate phosphoribosyl transferase are sensitive to IRI, the latter reversed by boosting NAD⁺ content by NAM.⁷⁷ Third, in patients with AKI, urinary quinolinate levels associate with AKI, while NAM lessens AKI in patients undergoing cardiac surgery.⁷⁷

While impairment in the *de novo* NAD⁺ biosynthesis pathway is generally most prominent in AKI, impairment in the other NAD⁺ biosynthesis pathways also occurs. In other AKI models, NAD⁺ synthesis is also impaired, and boosting NAD⁺ is generally protective.^{73–76}

Certain enzymes avidly consume NAD⁺ when activated, a classic example being poly (ADP-ribose)

polymerase (PARP). These enzymes are needed for DNA repair and endoplasmic reticulum stress responses. Activated in AKI, these enzymes avidly use NAD⁺. Inhibition of such activity restores NAD⁺ and protects against AKI. Other NAD⁺-consuming enzymes include sirtuins.

In addition to NAD⁺ depletion *per se*, interest centers on whether metabolites that accumulate or become insufficient in NAD⁺ metabolism are also functionally significant in AKI.⁷⁶

Diabetes

Age-dependent occurrence of diabetes may predispose to age-dependent AKI. Diabetes promotes renal senescence through telomere shortening, inflammation, mitochondrial dysfunction, p21^{Cip1} and p16^{Ink4a} induction, and loss of α -Klotho, among other mechanisms.^{78–80}

Sirtuins

Sirtuins decrease with age.⁸¹ Sirtuins repair DNA, exert salutary effects on mitochondria and cellular metabolism, and are anti-inflammatory and antioxidant. Sirtuins, by their deacetylation actions, use NAD⁺, and this effect *per se* may worsen AKI. Yet, several sirtuins protect against AKI because of the intrinsic salutary effects of sirtuins, independent of changes in NAD⁺. For example, kidney-specific overexpression of Sirt1, expected to deplete NAD⁺, protects against nephrotoxic AKI by augmenting oxidant scavenging.⁸² The kidney of younger mice compared with older mice exhibit higher Sirt1 expression, and this underlies the relative resistance of younger mice to AKI.⁸³ Older mice display not only a decrease in Sirt1 but also a decrease in NAD⁺, and the age-dependent AKI susceptibility can be reversed by boosting NAD⁺; such ameliorative effects, however, are dependent on Sirt1.⁸⁴ Analogous findings regarding Sirt3 and NAD⁺-boosting pertain to sepsis-induced AKI.⁸⁵ Thus, there is a bidirectionality in that

Sirt1 and Sirt3, as other sirtuins, consume NAD⁺, but the beneficial effects of NAD⁺-boosting require functional sirtuins in these settings.

Mitochondrial Injury

Loss of ATP and an inability to generate ATP—a consequence of mitochondrial injury—are key determinants of AKI and its outcomes.⁸⁶ Mitochondrial injury also drives AKI because such injury decreases NAD⁺, promotes oxidative stress, increases inflammation through the release of DAMPs, and causes cell death through the release of caspase 3 and mitochondrial DNA.⁸⁷ Mitochondria undergo a continual cycle of fusion and fission, with mitogenesis critically dependent on PGC-1 α ; these processes maintain an adequate number of healthy mitochondria.⁸⁷ Age impairs PGC-1 α expression,⁸⁸ mitochondrial biogenesis, and the mitochondrial cycle of fusion and fission. Age-associated impairment in mitochondrial integrity and homeostasis underlie the age-dependent susceptibility to AKI.^{87,89}

Autophagy

Autophagy cordons off injured organelles and macromolecules from cells, thereby shielding otherwise viable cells from the damaging effects of these injured structures.⁹⁰ A special form of autophagy—mitophagy—removes injured mitochondria, the latter especially threatening to cell viability. The relevance of both autophagy and mitophagy to AKI is that age impairs both processes.

Impaired autophagy and mitochondrial integrity may concomitantly occur in senescence because of reduced levels of MondoA, an antisenesence molecule that decreases with age.⁹¹ MondoA promotes both autophagy and mitochondrial integrity, the latter through Prdx3; indeed, mice deficient in MondoA exhibit exaggerated senescence after IRI.

Epigenetic Changes and AKI

Occurring with age, epigenetic changes include DNA methylation, histone modifications, and noncoding RNA.⁹² Such changes in the kidney and peripheral blood mononuclear cells correlate with age-associated loss of kidney function.^{93,94} Epigenetic changes are implicated in AKI.⁹² Age-induced epigenetic changes may thus predispose to AKI and constitute a form of “genetic memory” that determines the sensitivity to subsequent insults and outcomes after AKI.⁹²

Telomere Shortening

Telomere shortening occurs with age and predisposes to AKI⁹⁵ and CKD.⁹⁶ Telomere shortening attends each cell division, and such shortening may be hastened by assorted insults.^{95,96} In kidney transplant recipients, telomere length predicts, along with donor age and p16^{Ink4a} expression, the long-term kidney allograft function.⁹⁷ In kidney transplant recipients, delayed graft function associates with shorter telomere length.⁹⁸ Mice deficient in enzymes that maintain the integrity and length of telomere show worse kidney injury and delayed recovery from AKI.^{99,100} Thus, telomere

shortening, as occurs in aging, may predispose toward AKI, and, in turn, kidney injury may further damage and shorten telomeres, thereby exacerbating AKI.

Impaired Klotho Content

α -Klotho, a protein that promotes health span, decreases with age.¹⁰¹ The fact that Klotho^{+/-} mice are more susceptible to AKI, whereas α -Klotho-overexpressing mice are resistant to AKI, along with the finding that recombinant α -Klotho protects against AKI, raises the possibility that the age-associated decline in α -Klotho expression may contribute to increased susceptibility to AKI with age.¹⁰² Among other salutary characteristics including fostering repair, α -Klotho is antisenescent, anti-inflammatory, antioxidant, and antiapoptotic.^{102,103}

Decreased Expression of Other Nephroprotective Pathways

The risk for and the severity of AKI are mitigated by several inducible nephroprotective species including heme oxygenase-1^{104,105} and its upstream transcription factor Nrf2, the inducibility/expression of either decreasing with age.¹⁰⁶

Immunosenescence

Whatever the cause for AKI, renal inflammatory processes are involved in AKI, and exaggerated inflammatory responses contribute to worse AKI in old mice.

Immunosenescence reflects age-induced inflammaging and alterations in innate and adaptive immunity.^{107–109} Inflammaging originates from increased systemic levels of proinflammatory cytokines and induction of proinflammatory genes in the kidney and elsewhere. This proinflammatory milieu is driven by sources such as innate immunity-centered processes, T-cell aging, NF- κ B activation, senescent cells and SASP factors, unrestrained inflammasome activity, and gut microbiota dysbiosis.^{107–109}

Aging also leads to the accumulation of tertiary lymphoid structures in the kidney and other nonlymphoid organs.^{107–109} These structures congregate lymphocytes and proinflammatory fibroblasts, which incite self-amplifying inflammatory processes; these processes instigate autoimmune injury to renal parenchymal cells and cause fibrosis. Tertiary lymphoid structures enable a dialogue between senescence-associated T cells and age-associated B cells communicated through CD153/CD30 signaling.¹¹⁰ The interruption of such signaling reduces T-cell induction and, in turn, the B-cell population, with reduced inflammation and fibrosis. Interruption of such signaling in old mice, for example, aids AKI recovery.¹¹⁰

p21^{Cip1}

Age increases renal p21^{Cip1} expression.¹¹¹ Following IRI or cisplatin, p21^{-/-} mice exhibit worse AKI.^{112,113} Such findings may seem incongruent with a role for p21^{Cip1} in AKI-induced senescence. However, p21^{Cip1} is pleotropic with recognized antiapoptotic effects,²⁴ the latter implicated in its nephroprotection in AKI.¹¹⁴

Table 4. Clinical trials in AKI and other conditions

Therapy/Drug	Disease Setting	Patients	Trial Number	Completion
Aging-related clinical trials				
D+Q	Age-related dementia	Approximately 120	NCT04685590	2025
Metformin (TAME trial)	Aging and longevity	Approximately 3000	NCT02432287	2026
Fisetin	Frailty and inflammation in women	Approximately 40	NCT03430037	2025
Senolytic peptide UBX1325	Diabetic macular edema	Approximately 60	NCT04857996	2025
AKI and sepsis-related clinical trials				
VITAKI (NAM, vitamin B3)	AKI in septic shock	310	NCT04589546	2025
STOP-sepsis (fisetin)	Sepsis in older adults	220	NCT05758246	2026
AKI and cardiac surgery-related clinical trials				
NACAM (NAM)	Cardiac surgery-associated myocardial injury	304	NCT04750616	2025
AKI-RCT (NAM riboside)	Kidney function in cardiac surgery	100	NCT06521307	2026

D+Q, dasatinib+quercetin; NAM, nicotinamide.

Senescence and AKI: Therapeutic Strategies

General strategies that remove senescent cells or combat the consequences of senescence have been recently and comprehensively discussed.¹¹⁵ The following adopts this outline with regard to such strategies targeted to the kidney and AKI.

Clinical trials are currently examining the effect of senolytics or boosting NAD⁺ in human AKI and other diseases (Table 4).

Senescent cell killing relies on strategies that interrupt survival mechanisms these cells recruit. First, prosurvival pathways, such as Bcl-2, Bcl-xL, and Bcl-w, are upregulated in senescent cells, and thus, agents that target these pathways, *e.g.*, ABT-263, are senolytic; ABT-263 ameliorates AKI.³³ Second, senescent cells have high SA- β -Gal activity, and such augmented activity can be used to activate a prodrug within senescent cells, thereby restricting its cytotoxicity to these cells. Such an approach has been used with gemcitabine, a chemotherapeutic agent that blocks DNA synthesis; this strategy kills renal senescent cells and protects against the AKI to CKD transition.¹¹⁶ Third, senescent cells often exhibit increased p53 expression, the latter capable of inducing one of two fates: senescence or apoptosis.¹¹⁷ High-p53-expressing cells may be channeled to senescence because they concomitantly have increased expression of FOXO4 (among other survival factors), which, by binding p53, promote the nuclear pro-senescent effects of p53, rather than its proapoptotic effect. Designer peptides that interrupt the binding of FOXO4 and p53 force the egress of p53 into the cytoplasm; there, p53 triggers mitochondrial-dependent apoptosis of senescent cells. Such inhibitors reduce the loss of kidney function in aged mice.¹¹⁷

Fourth, relevant to the kidney are strategies targeting kidney-type glutaminase. This enzyme is essential for renal ammoniogenesis, which enables urinary net acid excretion. However, excessive ammoniogenesis promotes progressive tubulointerstitial disease,¹¹⁸ the latter recognized, in part, as a senescence process.¹¹⁹ In senescent cells, lysosomal defects leak protons in the cytoplasm, and such leakage activates glutaminase and stimulates ammoniogenesis¹²⁰; increased amounts of ammonia bind protons and thus

prevent intracellular acidosis and attendant death of senescent cells.¹²⁰ Inhibitors of glutaminase prevent ammonia production, thereby leading to acidosis-driven killing of senescent cells; such inhibitors improve kidney function and reduce histologic injury in aged mice.¹²⁰ Fifth, natural products such as the flavanoid, fisetin, may interrupt multiple, upregulated antiapoptotic pathways in senescent cells¹²¹; fisetin protects against IRI,¹²² as does another flavanoid quercetin.¹²³

Clearance of senescent cells by immune-based strategies include immune check point inhibitors (senescent cells may upregulate checkpoint proteins)^{115,124} and the targeting of specific antigens on senescent cells by CAR-T, vaccines, or antibody-drug conjugates.^{115,124} Senescent cell clearance by immune-based strategies are yet to be studied in AKI.

Senomorphics inhibit components of the SASP, and in models of CKD, senomorphics reduce kidney injury.¹²⁴ Both metformin and rapamycin are senomorphics and, at least in some studies, may protect against AKI.^{125,126} SASP factors, *e.g.*, signaling species (NF- κ B,¹²⁷ JAK/STAT,¹²⁸ cGAS-STING¹²⁹), specific cytokines (IL-6, TNF- α , MCP-1),¹³⁰ and inflammatory vasoconstrictors (endothelin-1),¹³¹ among others, are involved in AKI. Novel therapeutic approaches are being applied to SASP components. For example, using the proteolysis-targeting chimeras approach to impair the cGAS-STING pathway protects against AKI.¹³²

Senescence and AKI: Conclusion

The rapidly developing field of senotherapeutics, including second-generation senotherapies, and the encouraging findings from preclinical AKI studies raise the prospect for novel therapies for human AKI.^{115,124} Senolytics have now entered clinical trials in human disease, including AKI (Table 4).

Senescence is a dynamic field that continues to elucidate not just the pathogenic role of senescence in assorted diseases but also its physiologic and other salutary roles in embryology, organ development, wound healing, and cancer prevention.¹³³ The study of senescence also continues to uncover the complexities of senescent cells, in

all their heterogeneity, nuanced behavior, ecosystem-dependent characteristics, multifarious interactions with the immune system, and their potential for phenotype reversibility and reprogrammability.¹³⁴ Scientific discoveries in fast-moving fields have an uncanny way of disclosing serendipitous therapeutic insights; hopefully, such is the case regarding senescence and the introduction of novel therapies for AKI.

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Disclosure forms, as provided by each author, are available with the online version of the article at <http://links.lww.com/JSN/F603>.

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